mask.

Your money is *nobody else's* business.

Private invoices and checkout for businesses and AI agents, built on Midnight.

MIDNIGHT BUILDATHON

MASK-MIDNIGHT.VERCEL.APP



THE PROBLEM

A public ledger is a *postcard*. Anyone can read it.

01

A studio paying its team

Every salary, to the cent

- ✕ 14 wallets
- ✕ 14 amounts
- ✕ every month

02

A shop paying suppliers

Your margins, on display

- ✕ who you buy from
- ✕ what you pay
- ✕ when you reorder

03

A donor giving to a cause

A permanent, public link

- ✕ your wallet
- ✕ the cause
- ✕ the amount

THE IDEA

Public *proof*. Private payment.

WHAT THE CHAIN LEARNS

- That an invoice commitment exists
- Its kind, asset and expiry
- Open, settled or cancelled
- How many invoices have been paid
- A revenue claim's threshold, and that it holds

WHAT STAYS WITH THE PARTIES

- Merchant and payer identity
- Amount, title and memo
- Which payer settled which invoice
- The receipts and the revenue behind a claim
- The owner secret that can close or cancel

HOW IT WORKS

One hash. *No secrets.*

01

Commit

The merchant's browser hashes merchant key, amount, terms and a 256-bit salt into one commitment. Only the commitment goes on chain.

02

Pay

The payer reproduces the commitment from the real terms in the link. That proves they know them without publishing them. A shielded coin moves in the same transaction.

03

Settle

A nullifier refuses a second payment and a receipt is published. Amounts and parties never appear.

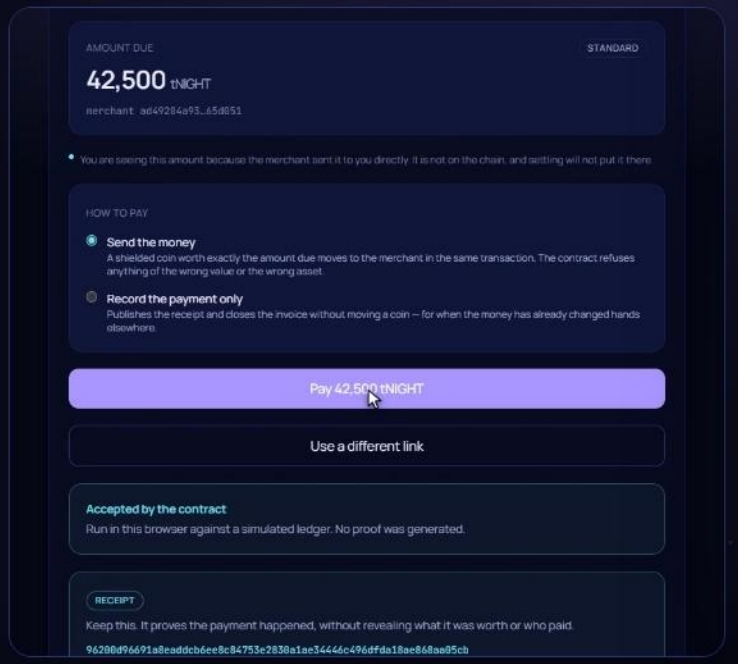
terms travel in the URL fragment, which browsers never send to a server

THE PRODUCT

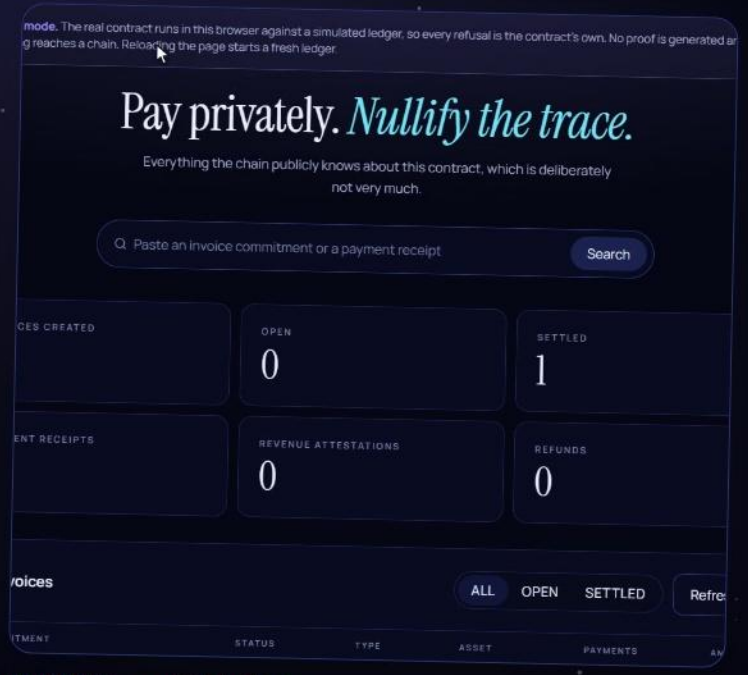
Create. Pay. *Verify.*



MERCHANT · CREATE



PAYER · RECEIPT



ANYONE · EXPLORER

Merchant and payer apps, an Explorer anyone can read without a wallet, a dashboard that follows the chain, refunds, cancel and close.

BEYOND INVOICES

Everything a *checkout* needs.

Standard, multi-pay, donation

One payment, many payers until closed, or an open collection.

tNIGHT, mUSDC, mUSAD

Settle in the network token or two stable assets.

Batch pay

Four invoices, one proof, one transaction.

Gift cards

Bearer cards: whoever holds the secret can spend it.

Mask Card

PIN and spend limit, no wallet at the till.

Revenue proofs

Prove at least X was earned, reveal nothing else.

Refunds

Once per payment, authorised by the owner secret.

Verify

Anyone checks a receipt, invoice or attestation.

MIDNIGHT INTEGRATION

Private state. *Public proof.*

YOUR MACHINE

- `merchant_key()`
- `invoice_salt()`
- `invoice_amount()`
- `payer_key()`
- `owner_secret()`

Witnesses. Never transmitted. A compile error if one reaches the ledger without `disclose()`.

ZK PROOF
PLONK

THE CHAIN

`invoices`	Map
`spent`	Set
`receipts`	Set
`receipt_tree`	HistoricMerkleTree
`attestations`	Set
`refunded`	Set
`created`	Counter

CONTRACTS

`mask_pay`
`mask_token`
`mask_card`
`mask_gift`

Compact 0.31.1, runtime 0.16.0, midnight-js 4.1.1, ledger v8

kind, asset and expiry are public on purpose · the amount is a witness no circuit discloses · the owner secret becomes a one-way tag

QUALITY ASSURANCE

Tested like a *judge* would.

155
contract tests
simulator, weighted to refusals

56
browser tests
desktop and phone viewport

56
backend tests
checkout sessions, webhooks

56
bot tests
public state only

19
SDK tests
Node SDK

PRIVACY ASSERTED, NOT ARGUED

The contract's public transcript for every call, indexer data and ledger state, session-store keys and webhook bodies. Tests prove their search sighted something first, so they can fail.

ON LIVE NETWORKS

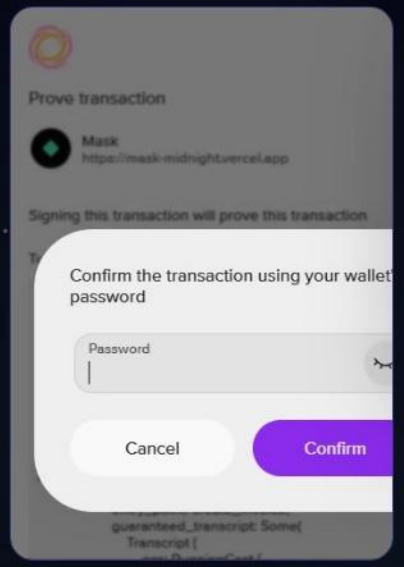
Midnight Preview and Preprod
13 of 13 circuit keys on chain
7 of 7 end-to-end checks passed

Real wallet. *Real chain.*

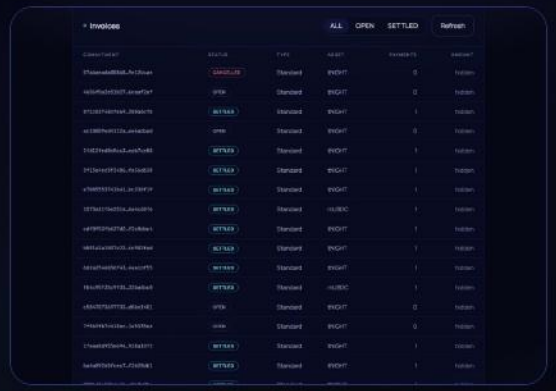
create_invoice
48 s
proved and confirmed through Lace

settle_invoice
67 s
a shielded mUSDC coin, receipt published

batch of four
100 s
four payments, one proof, one transaction



LACE · PROVE TRANSACTION



EXPLORER AFTER THE SESSION

Lace 2.4.2 on the deployed app, one continuous session: create, a shielded mUSDC payment, a replay refused, a refund, four invoices paid in one batch proof, a revenue proof, an owner cancel and a payment to the cancelled invoice refused. Each write proved and confirmed on Midnight Preview. The Explorer afterwards shows 33 invoices, 24 settled, 2 refunds, 2 revenue proofs.

Not on Preview: gift cards and Mask Card. Their contracts are not deployed there yet, so those clips are labelled demo mode.

Evidence in deployments/preview.lace-evidence.json

DEVELOPERS AND AGENTS

Build it into *your product*.

Node SDK

`@mask/node`

Invoices, links, webhooks, chain reads.

Python SDK

`mask-midnight`

Links, terms, webhook verification.

CLI

`@mask/cli`

Keys, invoices and on-chain checks.

MCP server

`@mask/mcp`

An AI agent creates and verifies over stdio.

Checkout backend

`sessions + webhooks`

No amount ever reaches your server.

Telegram bot

`public state only`

Looks up, watches settlement, holds no secret.

the one rule: your server never sees an amount. terms travel from browser to browser in a URL fragment.

AUDIENCE AND ADOPTION

Who it is for, *and how it spreads.*

WHO PAYS FOR PRIVACY

Freelancers and agencies
invoice in crypto without publishing their client list and rates

Payroll and supplier runs
pay four at once, prove totals to an accountant only

DAOs and grants
pay contributors, give auditors a proof, not a ledger

AI agents
buy services and settle invoices through an MCP server

ADOPTION PATH

1. **Try it with no wallet**
   The real contract runs in the browser: a judge or a customer feels it in a minute.
2. **Drop it into a checkout**
   SDKs, sessions and webhooks keep amounts off the merchant's server.
3. **Go on chain with Lace**
   Preview today, Preprod deployed.
4. **Prove, selectively**
   Revenue proofs turn private sales into something a lender can check.

STATED PLAINLY

What it *doesn't* do yet.

Timing correlation is not solved
Mitigated by volume, not cryptography.

Proving keys are 161 MB
A public deployment needs a CDN and per-circuit lazy fetching.

Revenue claims cover four payments
Circuits cannot loop; wider claims mean a wider variant.

No hosted prover
A prover would see what it exists to hide.

NEXT

01 Serve proving keys from a CDN, one circuit at a time

02 Compile wider revenue-claim variants

03 Deploy gift cards and Mask Card to Preview, so they run through Lace too

PROGRESS AND LINEAGE

What this *repository* adds.

BUILT AND VERIFIED IN THIS BUILD

- Deployed to Midnight Preview and Preprod, 13 of 13 keys, 7 of 7 live checks
- Real Lace wallet: create and record-only payment confirmed on chain, twice
- Revenue proofs wired end to end in the app, with refusals tested
- Dashboard reads settlement from the chain; payment codes for refunds and proofs
- Expiry in seconds, owner-secret cancel, refunds, privacy tests that can fail
- A film and this deck, recorded from the running product

LINEAGE

Mask began as AleoZKPay, a private-invoice product on Aleo by the same team.

Its flows, page structure and feature set are the source. The Compact contracts, circuits, private-state design, proving pipeline and interface are new work on Midnight, and the README says so.

mask.

Try it. *Break it.*

LIVE APP	mask-midnight.vercel.app
SOURCE	github.com/Pratiikpy/mask-midnight
LICENCE	Apache-2.0
NETWORK	Midnight Preview · Preprod

